



How latency, action modality and display modality influence the sense of agency: a virtual reality study

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Abstract

The objective of this study was to investigate the influences of latency (i.e., technical system response time), action modality (button press, voice command) and display modality (head-mounted display, monitor) on the sense of agency (SOA). SOA is the experience of controlling one's own actions and their corresponding effects in the environment. The $N=31$ (48% female, with a mean age of 24) participants had to interact repeatedly with three different objects (lamp, tablet and computer) in a virtual environment (presented on a monitor or via a head-mounted display) by using a voice command or pressing a button to turn the objects on. The objects reacted after a specific technical system response delay (150, 450 and 750 ms). Results showed that the SOA was weaker for actions employing voice commands opposed to button presses, except for the explicit SOA in the monitor condition. Higher latencies diminished the explicit, but not the implicit SOA. Neither the explicit nor the implicit SOA was significantly affected by the display modality. The findings in part support the weighting process of different agency cues of the underlying framework, and we propose to extend this model by a sense of presence. Users seem to react as if they have the impression that they are not able to control the technical system properly if they interact through a voice command. Therefore, human–computer interface designers could take account of our findings regarding the modality of an action by providing additional feedback cues to increase the SOA for interactions with voice interfaces.

Keywords Virtual reality · Sense of agency · Voice command · Latency · HMD · Human–computer interaction

1 Introduction

When a human picks up a pen, it seems reasonable that the person's mind commands the body to stoop, reach out and grab the pen. Therefore, the human is in control of that motion or, in other words, experiences agency. But what if the mind commands an action that is carried out by another body? For example, in a virtual reality (VR), the individual is represented by and acts through an avatar.

The question of particular interest to us is: Do we claim agency over the actions carried out by a surrogate, even though one can argue that the avatar is the one who executes the action? The objective of this research was to investigate the influences of latency, action modality and

display modality on the sense of agency within a virtual environment.

1.1 The sense of agency

According to Farrer et al. (2013), the term “sense of agency” (SOA) refers to the subjective experience of controlling one's own actions and the corresponding outcomes. Synofzik et al. (2008b) subdivided the SOA into first, the feeling of agency that is prevalent in our everyday experiences (Synofzik et al. 2009), and second, the judgement of agency. The feeling of agency refers to the implicit, low-level feeling of being an agent. On this level, attribution to the self is not really possible and an action can only be classified as not self-caused or self-caused (Synofzik et al. 2008a). Building up on the feeling of agency, the judgement of agency is a second belief-like level of agency, which is described as explicit and conceptual and offers the means to interpret the current context (Synofzik et al. 2008b). In contrast to the feeling of agency, the person can now make a clear statement about the authorship of the action, i.e., it is now possible to

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state by what or whom the action was initiated. Additionally, the SOA is not a simple distinction between yes and no, but rather runs along a non-discrete continuum or, in other words, one can experience weaker or stronger agency. Therefore, we have to reformulate our initial question: To what extent do we claim agency over the actions carried out by an avatar? To answer this question, it is necessary to investigate the SOA in a VR setting, which would be especially interesting if one forgets that he is in a VR (see Sect. 1.3 empirical evidence of the sense of agency).

To call ourselves the agent of an action, it is necessary to discriminate between internally and externally produced effects. The comparator model (Blakemore et al. 2002) describes mechanisms that are important to allow the claim of agency over an action. For this purpose, three different comparators are necessary, which compare several states (desired state = input to the motor system, estimated actual state = output of the motor system, predicted state = prognosis of the outcome of one's action). The sense of agency arises if the states are congruent.

Synofzik et al. (2013) integrate and extend the comparator model in their recent framework to cover the discrimination between feeling and judgement of agency. The model weights several agency cues, including sensory input and information about environment, on the cognitive, affective and sensorimotor levels. For example, you have the intention to press a switch (an internal cue on the sensorimotor level) to turn on a light. You see your hand pressing this switch and you feel the haptic feedback of this button press (both external cues on the sensorimotor level). Afterward, you see the light turning on (an external cue on the cognitive level) and have the knowledge that nobody else could have turned on the light (an internal cue based on external cues on the cognitive level). As a result, you should experience a strong agency over turning on the light because your brain integrates all those different cues. For this weighting process, which is based on Bayesian optimal cue integration, the availability and reliability of each cue are especially important. If a cue is highly variant, the reliability of this cue should be low. On the other hand, a cue is reliable, if strongly salient and clear-cut (Synofzik et al. 2013). As a result, the lack of a cue should lead to a diminished SOA.

1.2 Operationalization of the sense of agency

To capture the SOA in its entirety, it is necessary to respect the distinction between judgement and feeling of agency (Moore et al. 2012). Judgement of agency can be assessed through self-reports (Ebert and Wegner 2010; Berberian et al. 2012). For feeling of agency, there is a dispute if the phenomenon called “intentional binding” (IB) really captures the feeling of agency. To avoid this dispute (see Synofzik et al. 2009, 2013, for further details), the terms

“explicit” and “implicit” measures of the SOA are more appropriate than judgement and feeling of agency. Following this notion, we do not claim that intentional binding purely captures feeling of agency but can be used to implicitly measure SOA (Haggard et al. 2002). Intentional binding refers to a distorted temporal perception, in which the perceived action and the corresponding effect shift toward each other in time. According to Haggard et al. (2002), the longer the time delay between a voluntary action and effect, the weaker the implicit SOA. This phenomenon has proven to be robust and has been replicated in a number of studies (see Moore and Obhi 2012).

Engbert et al. (2007) proposed a fairly simple method—called interval estimation—to measure intentional binding. It consists of asking the participants to estimate the time between an action and the corresponding effect directly after an action. The shorter the interval is judged to be, the stronger the intentional binding and therefore the SOA. Both measurement methods—self-report and interval estimation—have the advantage that they can easily be employed within a virtual environment.

1.3 Empirical evidence of the sense of agency

Several factors have the potential to influence the SOA, including latency and action modality. Latency refers to the response time of a technical system, that is, the temporal delay between input into and output from a technical system. Several studies suggest that latency has a negative impact on the SOA (e.g., Haggard et al. 2002; Ebert and Wegner 2010; Kawabe 2013). Furthermore, the investigated range of the temporal delay should be considered. In the SOA literature, added latency typically varies between 0 and 1000 ms, often with three levels of added latencies, e.g., 100 versus 400 versus 700 ms (Ebert and Wegner 2010). These latencies seem rather high for simple computer tasks (e.g., office applications) but are in standard range for example in speech recognition software like “Google Smart Home” (Google LLC. 2019). It should be noted that latency is not only a factor for SOA but also a crucial factor for VR applications. Furthermore, in the context of SOA different latencies were, to the best of our knowledge, insufficiently investigated within a VR.

Based on the fact that speech interfaces are a naturalistic approach to interact in a VR, it is unfortunate that Limerick et al. (2015) did investigate the relationship between voice commands and SOA but not in a VR. In a computer experiment, they compared how a voice command (“go”) and a button press were related to implicit SOA. It is important to note, of course, that a voice command is a more continuous action (less clear onset and end) than a button press. Therefore, the experimenters had to agree on the distinct point, called the point of action, at which the action takes

place: at the beginning of the voice command or at the end. The results by Limerick et al. (2015) show that for both the point of action at the end and at the beginning of the command, the SOA tends to be stronger for interactions via button press as opposed to voice commands. However, their interesting findings should be confirmed by another temporal binding task and an additional explicit method (judgement of agency) which could provide further insight into the impact of action modality on the SOA. The model of Synofzik et al. (2013) provides a possible explanation for the diminished SOA in interaction with speech interfaces. It states that the SOA arises from the integration of several cues. The two actions (button press and voice command) differ mainly in the action feedback provided, e.g., seeing your hands reach out. This feedback is included as the sensory input into the model of Synofzik et al. (2013) and consists of auditory, haptic and visual cues. Limerick et al. (2015) noted that the sensory input cue for a button press is mainly visual and haptic. Here, the auditory component of the cue should have no influence, or only very little, on the SOA. For speech, however, the main cue is auditory, with visual feedback playing a negligible role. However, it has been reported that for the SOA the visual action feedback is dominant (Evans et al. 2015).

However, feedback of an action can not only be altered, but also suppressed. Such suppression leads to a lack of sensory feedback (e.g., visual) and therefore to a weaker SOA (Synofzik et al. 2006). A natural approach for suppressed feedback is the use of a different display modality. As a result of the representation via a head-mounted display (HMD), for instance, visual sensory feedback is limited to the scene, and no peripheral cues are integrated, e.g., seeing one's own hands and arms. According to Schoeffler et al. (2015), an HMD is a binocular device, which is worn on the head and has at least one integrated display. HMDs represent virtual realities or environments that are defined as simulated three-dimensional worlds, which the user not only perceives but also interacts with (Borsci et al. 2016). Thereby it is necessary to understand two basic concepts: immersion and the sense of presence (SOP), while immersion is dependent on things like frame rate, the given field of view, the realism of the displayed scene and latency. The SOP builds upon immersion in so far as without a certain degree of immersion it does not seem possible to reach a high degree in it. The concept therefore refers to the human response to the immersion of the environment (Slater et al. 2009). Wirth et al. (2007) developed their definition of SOP around the question which environment a person temporarily accepts as their reality, may it be virtual or real. The answer to this question is based upon two factors: (1) The person actually feels as if they were in the environment (self-location) and (2) has the ability to do something there (possible action). As Kiltner et al. (2012) stated, for a SOP to emerge it is

necessary to establish the feeling that you are the agent of that action in this virtual environment. The importance of interactivity with a VR implies the possibility of investigating the SOA in VR (Blom and Beckhaus 2014; Turchet 2015). Furthermore, Lee et al. (2013) point out that it is reasonable to investigate different verbal and haptic action modalities in alternate realities. An HMD has the advantage that it offers the ability to control and manipulate feedback of an action, rather than simply eliminating feedback. The knowledge that the visual action feedback for a button press is dominant, combined with the fact that there is a lack of visual action feedback while wearing an HMD, i.e., not seeing your hand pressing the mouse button, leads to the assumption that the difference in the SOA between button press and voice command should disappear while wearing an HMD.

1.4 Research question and hypotheses

The model by Synofzik et al. (2013) describes several cues that are essential for a SOA and, hence, provides a possible explanation for the weakened SOA in interactions employing a voice command. Therefore, our research focuses on the question: Is the limited visual feedback the cause for a diminished SOA when employing a voice command? For this purpose via employing a HMD, we manipulate the sensory input, especially the visual feedback of an action, through different action modalities and display modalities.

To investigate our research question, we specify the following three hypotheses:

H1 For higher latencies, the SOA should be weaker.

H2 For representation via head-mounted display, the SOA should be weaker as opposed to the representation by a conventional monitor.

H3a For actions employing a button press, the SOA should be greater as opposed to actions employing a voice command in the monitor condition.

H3b The differences regarding the SOA between a button press and a voice command should disappear in virtual reality.

2 Method

2.1 Participants

Thirty-two participants took part in the experiment. One participant had to be excluded from the analysis due to problems with pronunciation, which led to such poor recognition

by the software that the experiment had to be aborted. As a result, the final sample size which we analyzed was $N=31$. The participants were 20–35 years old ($M=23.81$, $SD=3.25$) and 15 participants were female (48%). In return for their participation, each participant was offered a 1-h course credit to satisfy an academic requirement. All of them gave informed consent at the beginning of the experiment.

With the help of G*Power 3, we estimated the required number of participants to identify a significant difference between the button press and the voice command based on the results by Limerick et al. (2015). G*Power suggested that at least 32 participants were necessary for the experiment ($\alpha=0.05$, $\text{power}= .90$).

2.2 Design

A $3 \times 2 \times 2$ within subject design was used with the following independent variables (IVs): (IV1) latency (0 ms, 300 ms and 600 ms), (IV2) display modality (monitor, HMD) and (IV3) action modality (button press, voice command). Based on tests prior to the experiment which showed a base latency of around 50 ms for button presses and 150 ms for voice command, 100 ms was added to every latency condition in the button press condition to balance the two different base latencies. Therefore, we will report the actual end-to-end latencies below, which are 150 ms, 450 ms and 750 ms.

2.3 Dependent measures

Following Kawabe (2013), the question for the explicit SOA was: “Referred to this action: Rate the extent to which your button press/voice command was able to causally control the object?”, on a response scale from 1 (“I did not control the object at all”) to 7 (“I fully controlled the object”). To assess the implicit SOA, the question was: “Referred to this action: How much time passed between your button press/voice command and when the object first started to react?”, following Engbert et al. (2008), on a scale with an open input from 1 to 1000 ms.

The SOP was captured via the MEC Spatial Presence Questionnaire (Vorderer et al. 2004). The MEC Spatial Presence Questionnaire (MEC-SPQ) is a questionnaire specifically designed to measure the SOP on a 5-point scale. According to Havranek et al. (2012), we only use the two 4-item scales spatial presence: self-location and spatial presence: possible actions from this questionnaire. To verify the reliability of both constructs, we calculated Cronbach’s alpha. For self-location, $\alpha = .789$. This value is above the proposed threshold of $\alpha = .7$ and therefore considered to be in bounds of a reliable score (Field 2009). Hence, we used the mean value for each participant across the four items for analysis. The measured reliability of spatial presence: possible actions, was $\alpha = .204$ and so below the necessary value.

After a detailed inspection of the items, we concluded that one item seemed inappropriate for our experiment. The item in question reads as follows: “I felt like I could move around among the objects in the presentation.” The subjects rated the item especially low because in our simulation it was not possible to move around in any condition. As a consequence, we excluded this particular item and compute the Cronbach’s $\alpha = .349$ again. Therefore, we treat the remaining three items of the possible actions scale separately.

2.4 Hardware and software deployed

The PC system on which the program was running consisted of the following components: An I7-5820k processor constructed by Intel Systems™, a NVIDIA GeForce 980 graphic card, 32 GB working memory and a SAMSUNG 850 EVO SSD hard drive. The operating system was Windows 7 Professional 64 bit. The peripherals used were a Logitech G303 gaming mouse with a refresh rate of 1000 Hz, and, to display the scene in the experimental condition without HMD, an Acer xB240HA 24-inch monitor with full-HD (1920×1080) resolution and a 144-Hz refresh rate. For the HMD condition, an Oculus Rift Development Kit 2 with full-HD (1920×1080) resolution, a 75-Hz refresh rate and a 100° field of view was employed.

We developed our VR scene in Unreal Engine v.4.10 (Epic Games, Inc.). Additionally, we used a CMUSphinx v.4 plugin to allow voice command input. The three-dimensional scene depicted a simple office room with artificial lighting. The user could see the scene through the first-person perspective of an avatar. This avatar was placed at the inner corner of an L-shaped table. The arrangement described is shown in Fig. 1.

The actor had three different interactive objects (lamp, tablet and desktop computer) which were placed on the table (see Fig. 2).

2.5 Procedure

Before the actual experiment, the participants had to take part in two practice blocks. The first one was designed to familiarize the participants with estimating short time intervals. Their task was to estimate the time of delay (ten randomized latencies ranging from 0 to 900 ms in steps of 100 ms) between their action and the effect, and afterward they were informed about the real latency.

The second practice block was to make the above-mentioned point of action clear to the participants. Throughout the experiment, the point of action was set at the end of the voice command. In this practice block, participants were instructed to speak the given voice command (“system”), while watching a virtual traffic light. The traffic light was set to red at the beginning of each trial. It switched to yellow

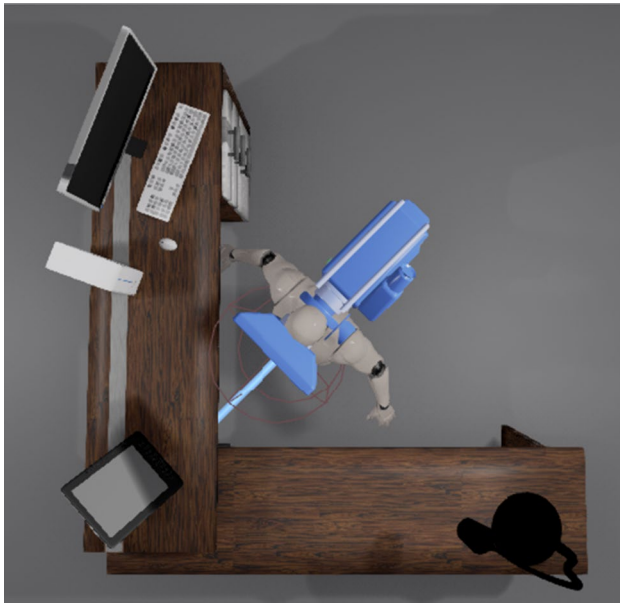


Fig. 1 Top-down view of the scene and the avatar

while the person spoke the word and shifted to green at the moment the system recognized the word. This was when the participants were supposed to press the left mouse button. They were given constant feedback about their action and the desired point of action, e.g., too early, too late. This procedure was repeated until they succeeded in acting within a set error range of 75 ms to the point of action five times in a row. This practice block was conducted with no added latency.

Fig. 2 f.l.t.r. Objects: lamp, tablet, computer with their states off and on



This training phase was followed by the experimental trial, which was divided into two blocks of trials (random order) differing in display modality. If the representation was via monitor, the participants looked around using the mouse, whereas if the representation was via HMD they used head movements. They were given the instruction to look around the scene and focus on one of the three objects. This object was highlighted in blue so that the participant had visual feedback of the correct fixation. An icon at the lower edge of the field of view indicated which action was possible: a “speaking symbol” for a voice command, or a hand symbol for a mouse button press. If the speech recognition did not correctly detect the voice command, the participants were asked to repeat the voice command. For a detailed depiction of the participants’ view, see Fig. 3.

We set up the program such that only a correctly highlighted object and the correct action corresponding to the icon displayed could produce an effect. The participants were free in their choice of object and the point in time chosen to execute the action, although they were instructed to interact with every object equally often, but not in any given order. One of three added latencies was applied randomly after the action was performed. After this short delay, the participants witnessed the effects of their actions. Then the dependent variables, including a reminder of the point of action, were queried in which the participants verbally answered (see Fig. 4). The experimental procedure was repeated 36 times for each block, for a total of 72 trials, so that the experimental task took around 25–30 min. After each block, the participants rated their SOP.

Fig. 3 First-person view with action symbols highlighted in yellow

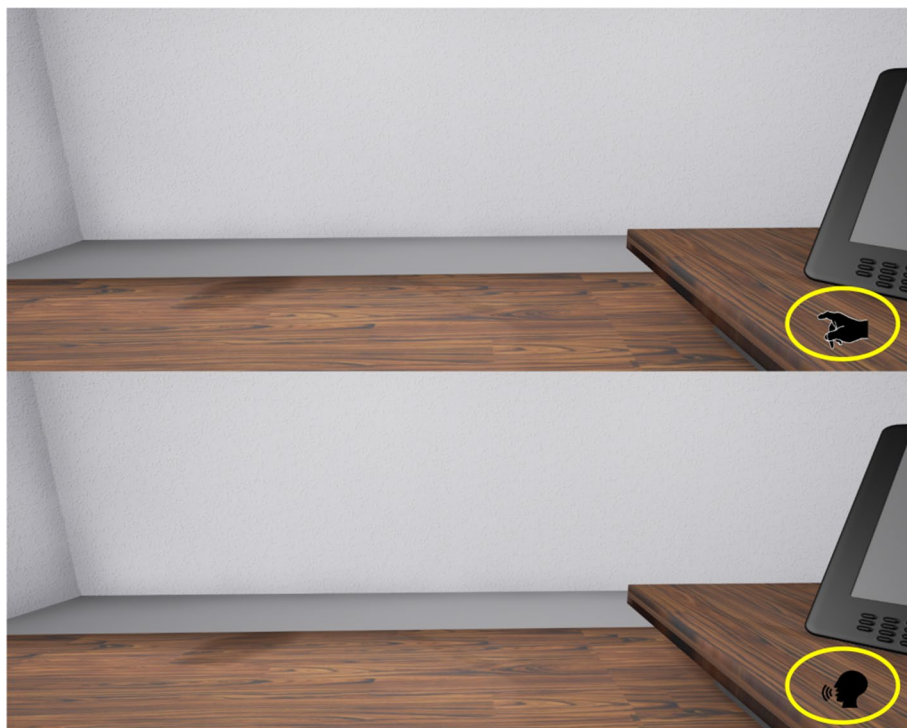
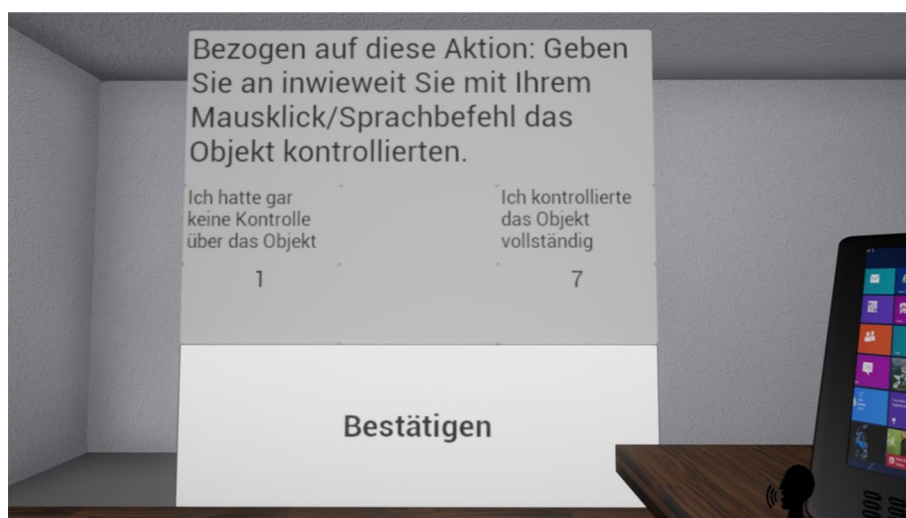


Fig. 4 Dependent variable panel (in this case for the explicit SOA)



Additionally, we implemented a routine to ensure that participants must consider the possibility that there could be another agent (e.g., the avatar, a confederate etc.) involved, because only with this possibility self-reports make sense (Moore et al. 2012) and opposed to most of the literature, our experiment lacked a passive or involuntary action condition (e.g., Engbert et al. 2007). This routine contains that in one out of five cases, so that there was a 20% chance in each trial that the objects (lamp, tablet and desktop computer) were able to switch their state without previous input. In such cases, only the last object that was interacted with was able to exhibit this behavior, 1 s after the dependent variables

were answered. The effect remained intact until the next interaction with this object, in order to increase the chance that a participant would perceive the random change. At the end of the experiment, the participants verbally answered two questions verifying our manipulation of another possible agent; if someone else controlled the objects and how often they noticed the change in the objects state without prior input (note that these questions were missing for 3 participants, hence $N=28$). A total of 43% of participants did not perceive another agent at all, while 29% of participants responded with a 2 on a five-point scale, indicating only a limited perception of another possible agent. In contrast,

only 11% of participants felt certain that another agent was involved. Furthermore, only 5.50 (SD=7.15) of an average of 14 uninitiated effects were perceived.

3 Results

Every statistical analysis was conducted on a 5% significance level (two-tailed). Every effect size was interpreted according to the conventions by Cohen (1992). For all statistical tests, we examined the assumption criteria and used the Greenhouse–Geiser correction for the *df*. We excluded every trial in which the action–effect relationship was not entirely clear (e.g., the program failed to recognize the command, the participants pressed the button twice). With this procedure, 1.21% of all trials were excluded.

To examine implicit SOA, we calculated the interval estimation error between actual end-to-end latencies and the estimated intervals according to Coyle et al. (2012). Thereby lower scores represent weaker implicit SOA and vice versa.

We also collected data about the three objects. The interaction count across all conditions was 821 for the tablet, 717 for the lamp and 708 for the computer. The following

patterns of results are consistent within the three objects as well as in the overall data.

For the analysis, we calculated the mean judgement scores for every participant in each condition across the objects (see Fig. 5) and the mean IB scores (which represent the implicit SOA) for every participant in each condition across the objects (see Fig. 6). To test our hypotheses, we employed two repeated measures analysis of variance (rmANOVA) with the three factors latency, display and action modality ($3 \times 2 \times 2$) one for explicit SOA and one for implicit SOA.

H1 Latency: longer latencies weakens the SOA

There was a significant large main effect of latency on the explicit SOA, $F(1.28, 38.48) = 30.50$, $\eta^2 = .50$ and on the implicit SOA, $F(1.26, 37.65) = 29.73$, $\eta^2 = .50$. In support of H1, the Bonferroni post hoc test revealed that the explicit SOA was significantly stronger in the 150 ms condition than in the 450 ms and the 750 ms conditions (both $p < .001$). Furthermore, the 450 ms condition scored significantly higher than the 750 ms condition ($p < .001$). In contrast to explicit SOA, the Bonferroni post hoc test revealed that the implicit SOA was significantly stronger

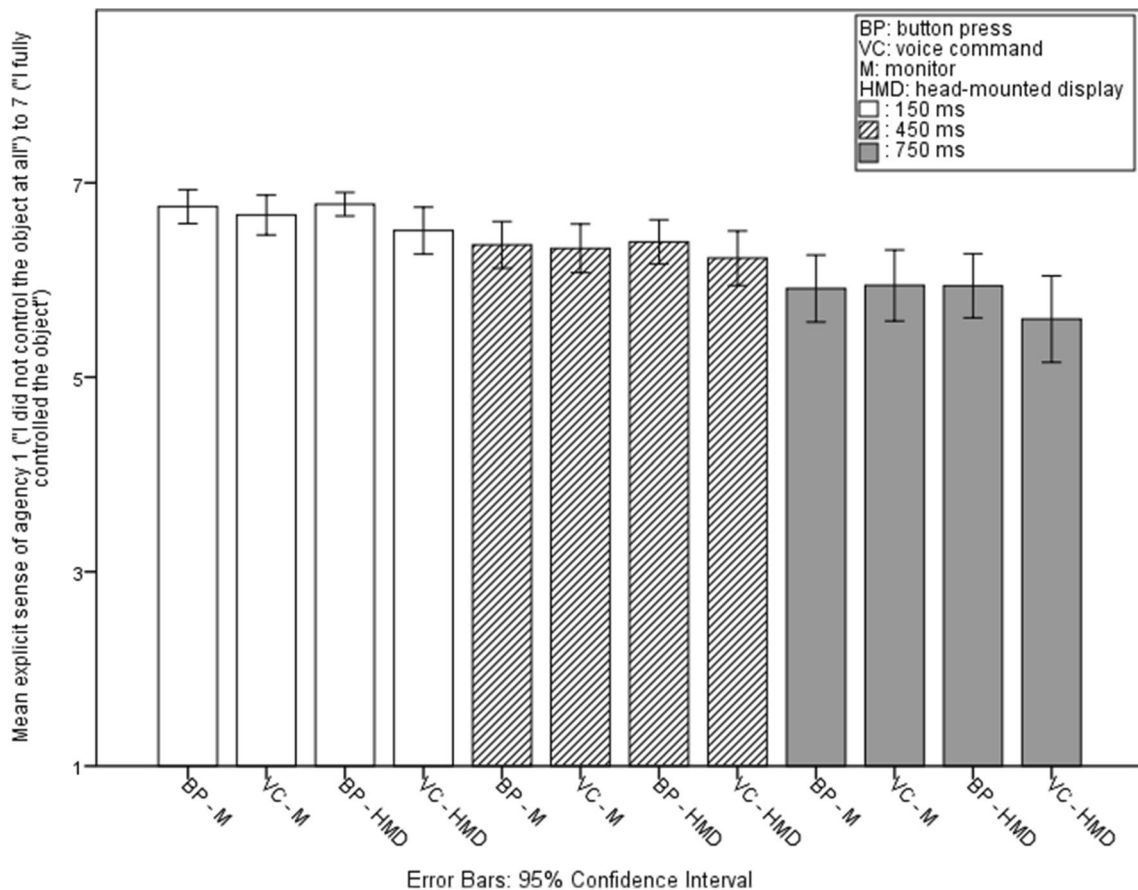


Fig. 5 Error bar diagram of the explicit SOA

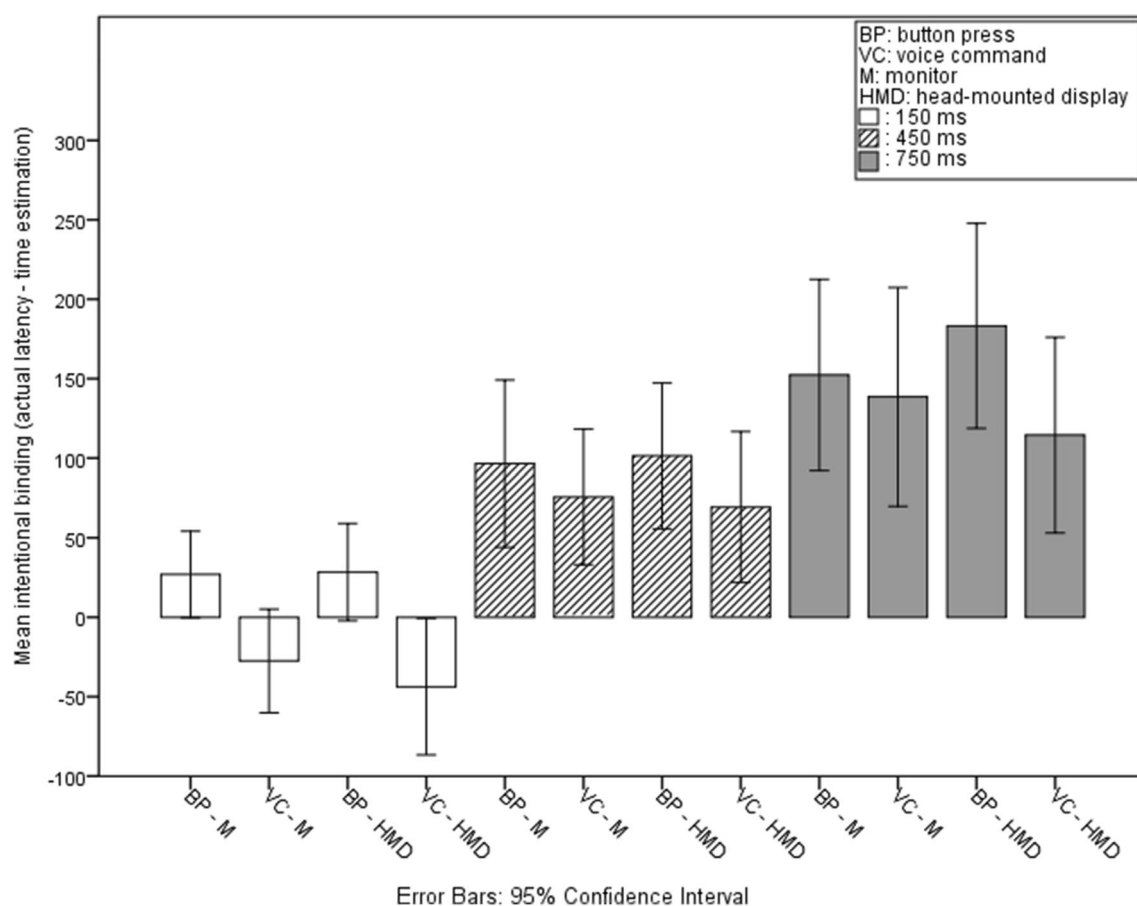


Fig. 6 Error bar diagram of the implicit SOA

in the 750 ms condition than in the 450 ms and the 150 ms conditions (both $p < .002$). Furthermore, IB was significantly stronger in the 450 ms condition than the 150 ms condition ($p < .001$). Therefore, we accept H1 on the explicit level of SOA but have to reject H1 on the implicit level of SOA.

H2 Display modality: in VR weakening of the SOA

For display modality, we found no significant main effect neither on the explicit SOA, $F(1, 30) = 0.99$, $p = .328$, $\eta^2 = .03$, nor on the implicit SOA, $F(1, 30) = 0.02$, $p = .900$, $\eta^2 = .001$. Therefore, we have to reject the hypothesis 2.

For the last hypotheses, we calculated separate rmANOVA with the two factors latency and action modality (3×2) within the display modalities: one for explicit SOA and one for implicit SOA.

H3a Action modality: actions employing a voice command weakens the SOA in the monitor condition

In the monitor condition, there was a no significant main effect of action modality on the explicit SOA, $F(1, 30) = .42$,

$p = .522$, $\eta^2 = .01$, but a significant large main effect on the implicit SOA, $F(1, 30) = 7.16$, $\eta^2 = .19$. As expected, the Bonferroni post hoc test revealed that the SOA was significantly stronger for a button press than for a voice command in the monitor condition (for implicit $p = .012$). Therefore, our results support our hypothesis H3a) only for the implicit SOA but not for the explicit SOA.

H3b Action modality: in VR no differences between button press and voice command for SOA

In VR, we found a significant large main effect of action modality on the explicit SOA, $F(1, 30) = 4.16$, $\eta^2 = .13$, as well as for the implicit SOA, $F(1, 30) = 20.43$, $\eta^2 = .41$. The Bonferroni post hoc test revealed that the SOA was significantly stronger for a button press than for voice command in VR (for explicit $p = .040$; for implicit $p < .001$). Therefore, we have to reject hypothesis 3b).

For spatial presence, the assumptions for the ANOVA were violated. Therefore, we conducted the Wilcoxon signed-rank test with the factor display modality. The results (see Table 1) show a large change when running

Table 1 Results of the Wilcoxon signed-rank test for SOP

Scale	Item	<i>Mdn</i> HMD	<i>Mdn</i> Monitor	<i>z</i>	<i>p</i>	<i>r</i>
Self-location	Items 1–4	3.75	2.00	−4.87	<0.001	−0.62
Possible action	Item 1	4.00	3.00	−2.75	=0.006	−0.35
	Item 3	3.00	3.00	−2.35	=0.019	−0.30
	Item 4	2.00	1.00	−3.38	=0.001	−0.43

the experiment with a HMD for the spatial presence: self-location and medium changes for spatial presence: possible action item 1, item 3 and item 4.

4 Discussion

The objective of our experiment was to investigate the differences in SOA between button press and voice command and examine the influence of the visual cues of these actions, based on the model of Synofzik et al. (2013). As Synofzik et al. (2008a) suggest, we measured both aspects of the SOA (explicit and implicit). To investigate the visual feedback systematically, we used a HMD and compared this with the monitor condition.

4.1 Latency

The results show that the SOA ratings confirm our expectations that high end-to-end latencies reduce the explicit SOA. This finding is in line with most of the literature (e.g., Ebert and Wegner 2010; Kawabe 2013; Kawabe et al. 2013). Furthermore, the theoretical frameworks predict this relationship between the SOA and the length of a time interval.

In contrast to our expectations and the explicit SOA, higher actual end-to-end latencies yielded stronger implicit SOA. This result is not in line with most of the literature (except for Ebert and Wegner 2010), where longer time intervals between action and effect impact implicit SOA negatively (e.g., Haggard et al. 2002; Engbert et al. 2008). This finding is quite surprising, and we are not entirely sure what cause underlies but we will offer some possible explanations. First, the time estimations were highly correlated with the actual end-to-end latencies, so it seems that the participants perceived the different levels of latency. We are able to exclude the possibility that a range bias has influenced our results because if we cluster our participants based on the variances of their time estimations into a low and high variance group the same patterns of results emerge in both groups. Additionally Ebert and Wegner (2010) found similar results, whereas the time estimations track the added latency accurately, the binding effect is strongest in the high-latency condition. This contributes to recent literature indicating problems with IB as a measurement of SOA (Synofzik et al. 2009), especially due to the small negative correlation

with self-reported SOA. One explanation of our findings is offered by Obhi and Hall (2011). According to them, if a computer is the potential cause of an effect, intentional binding will not occur while explicit ratings for agency will remain unaffected. This also applies to situations in which the participants are aware that they are the actor. However, their results indicate that no significant intentional binding occurs while we found significant binding but in the opposite direction. Therefore, Obhi and Halls (2011) findings are able to explain our results just in part. A further approach to better explain our unexpected findings would be the employment of a baseline condition including passive trials so the influence of SOA could be determined more accurately. Haggard et al. (2002) pursued a similar path to this approach by measuring SOA with the comparison of an action condition and a baseline condition. Although they used a different measurement method the similar procedure should be applicable when the time estimation method is employed.

4.2 Display modality

Neither explicit nor implicit SOA showed any significant differences as a function of the display modality. Therefore, we could not confirm that reduced visual cues of an action weaken the SOA, as the framework by Synofzik et al. (2013) would suggest. Thereby it should be noted that the display modalities differ not only in the available action feedback, but also in the evoked SOP. Our results show a higher SOP in the HMD condition and therefore it is most reasonable that the lack of visual sensory feedback has been compensated by the higher spatial presence. Thus, even though our results didn't support our hypothesis, the framework by Synofzik et al. (2013) can explain the findings if we factor the SOP.

4.3 Action modality

On the one hand, for the monitor condition only the results for implicit SOA show a stronger SOA for button presses than for voice commands. This finding does not match the results for explicit SOA. On the other hand, for the HMD condition both implicit and explicit SOA show a stronger SOA for button presses than for voice commands. This result is pretty straight forward as it would mean that we

can assume for button presses there are more cues which are more heavily weighted for a stronger SOA, which is in line with the framework by Synofzik et al. (2013), except for the explicit SOA in the monitor condition.

But if we dive deeper into the data, and look into the action modality separated by display condition as our hypotheses H3b suggests, the results become quite more complicated to interpret. Our initial hypothesis was that the difference between button press and voice command should disappear in VR because the HMD shuts down visual feedback of the action. This is not the case and if we inspect the effect sizes it becomes clear that the differences become even larger in the HMD condition. With the addition of SOP as a cue to the framework by Synofzik et al. (2013), we can explain the fact that the SOA strengthens for button presses in the HMD condition.

Why though the SOA for voice commands gets especially low in the HMD condition cannot be explained by the sense of presence. Our best explanation for this strange finding is the uncertainty of the participants based on their interaction with two new technologies (VR and speech interface) simultaneously. This could result in an increased arousal and as a consequence the participants had difficulties regarding the correct point of action. This would explain that the differences are especially high for the implicit measurement.

4.4 Limitations and future research

As a consequence of consciously omitting an adequate control condition, we had to make participants believe in another agent. Although we informed our participants of the possibility of another agent, evaluation of the questionnaire data indicates that they were blind to the uncontrolled effects and thus perceived no other agent. Apart from these questions of control, the preponderance of “7” ratings for the explicit SOA reinforces this conclusion. This minor imperfection has the potential to explain the non-significant results on display modality.

Due to the positioning of our objects, it was not possible to see all three at the same time and thus perceives uncontrolled reactions of these objects. The comparison between the HMD and monitor conditions could have been particularly influenced by this issue, because the HMD offered a smaller field of view. Accordingly, insecurity about agency could have been increased in the monitor condition. However, we are not able to conclude this with reasonable certainty because we did not ask the question after every block. We could have avoided this issue through some alterations, i.e., by rearranging the objects or asking the participants about the additional agent after every block.

Regarding the larger differences between button press and voice command in the HMD condition, our findings cannot easily be explained with our theoretical argumentation of

the weighing process between the sensory cues. We tried to find any other academic study that could at least in part point in the same direction as our results but we were not able to find any. Nonetheless, we are still confident in the general methodological aspect and the design of our study, with some minor alterations (e.g., including passive trials).

4.5 Implications and conclusion

Most of our findings support the importance of feedback cues and the assumed weighting process of these feedback cues as a function of their availability and reliability (Synofzik et al. 2013). With regard to the use of different action modalities in human–computer interaction, our findings suggest that users experience a weaker SOA in interactions with voice interfaces and even more so in VR applications. As a consequence thereof, users have the feeling that they are not fully in control of the technical system. In keeping with Shneiderman’s seventh rule of interface design, users wish a technical system to respond as intended so that they feel in charge of this system, i.e., experience agency (Shneiderman and Plaisant 2004). Our results reveal why users are unwilling to use speech interfaces (Aylett et al. 2014) in not so fail-safe tasks like calling a cab. One factor could be that voice commands are a less familiar interaction. With speech interfaces (e.g., Alexa, Siri, or Google Smart Home) gaining more prevalence in our daily lives the user should become more familiar with this technology and this could lead to more accurate results with a similar study in the future. Based on the practical implications of the model by Synofzik et al. (2013), designers could account for our findings by providing additional feedback cues to increase the SOA in interactions with voice interfaces. For example, an additional on-screen electronic audio visualizer could provide visual feedback on a voice command. Providing additional haptic cues for an action could be possible through skin-based feedback. Furthermore, for a stronger SOA engineers should avoid latencies above 450 ms. With the use of an HMD, we applied a new methodological approach to investigate the SOA. Our results demonstrate that, similar to a virtual body ownership (Ijsselstein et al. 2006), the framework by Synofzik et al. (2013) should be extended for VR with the integration of SOP into the model as it could be then better used with actions carried out by avatars in VR applications. Open for discussion is where exactly the SOP should be placed in the framework, on the cognitive or sensorimotor level or somewhere in between. The effect sizes of our results indicate that the SOA varies on the interplay between display and action modality. This means that if the SOA is crucial for the VR application, the employment of different input modalities, e.g., voice command and button press, should be carefully designed with the

target group in mind. Therefore, more research has to be pursued to better understand the underlying mechanisms that occur with the use of VR.

Author's contribution PW and PS designed the study, conducted the experiment and wrote as well as edited the manuscript in equal parts. NR and TF supported in study design and supported and reviewed manuscript preparation. JK supported the general research project and reviewed the manuscript.

Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

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